

Products

Outline

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Factorial



Definition

Introduction: Definition

- ▶ In this section we introduce the **product symbol** $\prod$, a compact notation for the product of many factors
- ▶ It plays for products the same role that the sum symbol $\sum$ plays for sums
- ▶ It is needed, for instance, to define the factorial and to work with determinants

Definition 1: product

Given n values $a_j \in \mathbb{R}$ with $j \in \{1, 2, \dots, n\}$, the **product**

$$a_1 a_2 \dots a_n$$

can be indicated in compact form with the product symbol:

$$\prod_{j=1}^n a_j$$

- ▶ The symbol j is called **product index**.
- ▶ The product $\prod_{j=1}^n a_j$ reads: “product for j from 1 to n of a_j ”.
- ▶ The product symbol is very useful when the terms a_j are explicitly defined as a function of the product index j .

Example

$$\prod_{j=1}^{10} \frac{1}{j} = 1 \frac{1}{2} \frac{1}{3} \frac{1}{4} \frac{1}{5} \frac{1}{6} \frac{1}{7} \frac{1}{8} \frac{1}{9} \frac{1}{10}$$

$$\prod_{j=3}^n j^2 = 3^2 4^2 5^2 \dots n^2$$

- ▶ The product index is a **dummy index**. This means that if j is replaced with i , k or any other index in all its occurrences the value of the product does not change.

Example: dummy index

- We have:

$$\prod_{j=1}^n j^2 = \prod_{i=1}^n i^2 = \prod_{k=1}^n k^2$$

as the three symbols indicate the product of the squares of the first n natural numbers (without zero).

- On the contrary, we have:

$$\prod_{j=1}^n j^2 \neq \prod_{j=1}^m j^2$$

as the two symbols indicate the product, respectively, of the first n and the first m squares of the natural numbers (without zero). Clearly, if $n \neq m$, the result is different.

Summary: Definition

Product symbol:

- ▶ $\prod_{j=1}^n a_j$ is a compact notation for $a_1 a_2 \cdots a_n$
- ▶ j is the product index; 1 and n are its first and last value

Dummy index:

- ▶ As for sums, renaming the index does not change the value of the product

Properties

Introduction: Properties

- ▶ We now explore six fundamental properties of products
- ▶ They parallel those of sums, with key differences due to multiplication: a constant factor appears as a **power** c^n
- ▶ The index properties (decomposition, translation, reflexion) are exactly the same as for sums

Observation 1: product with constant term

For every natural number $n \geq 1$ and real number c , we have:

$$\prod_{j=1}^n c = c^n \quad (1)$$

Proof

We have:

$$\begin{aligned}\prod_{j=1}^n c &= \underbrace{c \ c \ \dots \ c}_{c \text{ multiplied } n \text{ times}} \\ &= c^n\end{aligned}$$



Observation 2: product by a constant

For every product $\prod_{j=1}^n a_j$ and real number c , we have:

$$\prod_{j=1}^n (c a_j) = c^n \prod_{j=1}^n a_j \quad (2)$$

Proof

From the commutative and associative properties of the product, we have:

$$\begin{aligned}\prod_{j=1}^n (c \ a_j) &= c \ a_1 \ c \ a_2 \ \dots \ c \ a_n \\&= \underbrace{(c \ c \ \dots \ c)}_{c \text{ multiplied } n \text{ times}} \ (a_1 \ a_2 \ \dots \ a_n) \\&= c^n \prod_{j=1}^n a_j\end{aligned}$$



Observation 3: product of products

For every pair of products $\prod_{j=1}^n a_j$ and $\prod_{j=1}^n b_j$, we have:

$$\prod_{j=1}^n a_j \prod_{j=1}^n b_j = \prod_{j=1}^n (a_j b_j) \quad (3)$$

Proof

We have:

$$\begin{aligned}\prod_{j=1}^n a_j \prod_{j=1}^n b_j &= a_1 a_2 \dots a_n b_1 b_2 \dots b_n \\ &= a_1 b_1 a_2 b_2 \dots a_n b_n \\ &= \prod_{j=1}^n (a_j b_j)\end{aligned}$$



Observation 4: decomposition

For every product $\prod_{j=1}^{n+m} a_j$, we have:

$$\prod_{j=1}^{n+m} a_j = \prod_{j=1}^n a_j \prod_{j=n+1}^{n+m} a_j \quad (4)$$

Observation 5: index translation

For every product $\prod_{j=1}^n a_j$ and natural number $m \geq 1$, we have:

$$\prod_{j=1}^n a_j = \prod_{j=1+m}^{n+m} a_{j-m} = \prod_{j=1-m}^{n-m} a_{j+m} \quad (5)$$

Observation 6: index reflexion

For every product $\prod_{j=1}^n a_j$, we have:

$$\prod_{j=1}^n a_j = \prod_{j=1}^n a_{n-j+1} = \prod_{j=0}^{n-1} a_{n-j} \quad (6)$$

Proof

The three properties (decomposition, index translation and index reflexion) are simply different notations and/or arrangements of the terms in the products.



Summary: Properties (1/2)

Algebraic properties:

- ▶ Constant term: $\prod_{j=1}^n c = c^n$
- ▶ Product by a constant: $\prod_{j=1}^n (c a_j) = c^n \prod_{j=1}^n a_j$
(note the power c^n , not c as for sums)
- ▶ Product of products: $\prod_{j=1}^n a_j \prod_{j=1}^n b_j = \prod_{j=1}^n (a_j b_j)$

Summary: Properties (2/2)

Index properties:

- ▶ Decomposition: $\prod_{j=1}^{n+m} a_j = \prod_{j=1}^n a_j \prod_{j=n+1}^{n+m} a_j$
- ▶ Index translation: $\prod_{j=1}^n a_j = \prod_{j=1+m}^{n+m} a_{j-m}$
- ▶ Index reflexion: $\prod_{j=1}^n a_j = \prod_{j=1}^n a_{n-j+1}$

- ▶ They only rearrange or rename the factors of the product

Factorial

Introduction: Factorial

- ▶ The factorial is a fundamental operation in combinatorics and is defined using the product notation
- ▶ We give its definition and a useful formula for the ratio $\frac{n!}{(n-k)!}$
- ▶ This ratio appears, for example, when counting ordered selections of k objects among n

Definition 2: Factorial

The **factorial** of a natural number n , denoted as $n!$, is the product of the first n positive natural numbers. By convention, the factorial of 0 is defined as 1.

Formally, the factorial is defined as follows:

$$n! = \begin{cases} 1 & \text{if } n = 0, \\ \prod_{j=1}^n j & \text{if } n \geq 1. \end{cases} \quad (7)$$

Remarks:

- ▶ A closed formula for the factorial is not known but Stirling's approximation provides an asymptotic approximation.
- ▶ The factorial grows extremely fast: $10! = 3,628,800$

Observation 7

For every pair of natural numbers $n \geq 0$ and $k \in \{0, 1, \dots, n\}$, we have:

$$\frac{n!}{(n-k)!} = \prod_{j=1}^k (n-j+1) = \prod_{j=n-k+1}^n j \quad (8)$$

- Formula (8) is the product of k factors, from n and decreasing by one unit at a time.

Empty product

- ▶ By convention, a product with no factors (i.e., whose upper index is smaller than its lower index, as $\prod_{j=1}^0 a_j$) is equal to 1.
- ▶ This convention is consistent with $0! = 1$ and makes formula (8) valid also for $k = 0$ and $k = n$.

Proof

We have:

$$\begin{aligned}\frac{n!}{(n-k)!} &= \frac{\prod_{j=1}^n j}{\prod_{j=1}^{n-k} j} \\&= \frac{\prod_{j=1}^{n-k} j \prod_{j=n-k+1}^n j}{\prod_{j=1}^{n-k} j} \\&= \prod_{j=n-k+1}^n j \\&= \prod_{j=1}^k (n-j+1)\end{aligned}$$



Example

- Use of the observation:

$$\frac{100!}{98!} = \frac{100!}{(100-2)!} = \prod_{j=1}^2 (100 - j + 1) = 100 \cdot 99 = 9,900$$

Summary: Factorial

Factorial:

- ▶ $n! = \prod_{j=1}^n j$ for $n \geq 1$, and $0! = 1$ by convention
- ▶ An empty product is equal to 1

Ratio of factorials:

- ▶ $\frac{n!}{(n-k)!} = \prod_{j=n-k+1}^n j$: the product of k factors from n downwards
- ▶ Valid for every $k \in \{0, 1, \dots, n\}$

Key Takeaways

Summary (1/2)

1. Notation:

- ▶ The product symbol $\prod_{j=1}^n a_j$ provides compact notation for $a_1 a_2 \cdots a_n$
- ▶ The index is a dummy variable (like in sums)

2. Key Properties:

- ▶ Constant product: $\prod_{j=1}^n c = c^n$
- ▶ Factoring constants: $\prod_{j=1}^n (c a_j) = c^n \prod_{j=1}^n a_j$
- ▶ Index manipulation: decomposition, translation, and reflexion

Summary (2/2)

3. Factorial:

- ▶ $n! = \prod_{j=1}^n j$ for $n \geq 1$, and $0! = 1$ by convention
- ▶ Important formula: $\frac{n!}{(n-k)!} = \prod_{j=n-k+1}^n j$ (product of k successive integers)